PRERANA ACHAR

Intern at Max Planck Institute for Perceiving Systems

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EXPERIENCE

09/24 - Current **Max Planck Institute for Perceiving Systems - Tübingen, Germany** **Intern**
- Working with Dr. Michael J. Black's group on adding shoes and high heels motion to BEDLAM.

03/23 - 05/24 **Delft University of Technology - Delft, Netherlands** **Bachelor's Thesis**
- Working with Prof. Chirag Raman on the research project Material-Aware Multilayered Garment Draping for Digital Humans.
- Given a 3D human mesh, 3D garment(s) and garment material properties, the project aims to accurately drape multiple garments over the mesh and capture the physical properties and structural characteristics of each layer to achieve realistic clothing animations using graph neural networks

09/23 - 01/24 **Universidad Rey Juan Carlos (URJC) - Madrid, Spain** **Research Project**
- Working with Prof. Dan Casas on the research project ShoeMPL.
- This project aims to generate textured shoe offsets for the parametric body model SMPL from text inputs.

2022 **Text to Textile: Generation of Optimized 3D Garments from Textual Prompts**
- Independent research project
- This project is a text-to-3D clothing generator for digital humans that uses text prompts and template meshes that guide mesh deformations to generate clothing for fantasy avatars.
- Includes a blender addon to tweak parameters (length, texture, height of sleeve, skirt, jacket etc.) from user-entered prompts.

06/23 - 08/23 **JP Morgan Chase & Co.** **Intern**
- Abstract Syntax Tree and Language Parsing for an internal database language to Azure Databricks SQL.

07/22 - 08/22 **Net Tech Services India Pvt Ltd** **Intern**
- Improved a CNN-based model from 94% to 98% accuracy to extract handwritten data in English and Hindi from cheques.
- Designed graphics for ATM screens and cards

EDUCATION

10/20 - 06/24 **Dwarkadas J. Sanghvi College of Engineering** **B.Tech in IT (Honours)**
GPA - **9.42/10**
Courses: Mathematics I and II, Discrete Structures, Probability and Statistics, Statistical Analysis, Image Analysis and Computer Vision
Fourth Year Electives - Machine Learning and Advanced Machine Learning

2018 - 2020 **Chattrabhuj Narsee Monjee School - ISC Board** **Indian School Certificate**
Computer Science - **100%**
Overall - **96.2%**

2005 - 2018 **L.R.& S.M. Vissanji Academy - ICSE Board** **Indian Certificate of Secondary Education**
Computer Applications - **99%**
Overall - **93.5%**

AWARDS AND ACHIEVEMENTS

2022 **SIH National Winner** **Link to demo video**
Engineered a collaborative platform and mini-game builder in Unity to teach analytical concepts to students.
- Role: **Team Lead, Lead Programmer**
- Grant: **Rs. 100,000**

2022 **Project Ivory Flame presented at the Avishkar Research Convention**

LANGUAGES

English - Fluent
Kannada - Native
Hindi - Fluent
Tulu - Conversational
German - Basic
Marathi - Basic

SKILLS

Languages	Python, R, Java, Scala, C#, C++, Javascript, C, SQL, Go, Kotlin, Apache Pig
3D/Art Apps	Unity3D, Blender, Marvelous Designer, Android Studio, Autodesk Mudbox, Autodesk Maya, Krita, Gimp
IDEs	IntelliJ Idea, Eclipse, PyCharm, Visual Studio, VS Code
Other	Overleaf, Latex, Docker, MongoDB, PyTorch Lightning, Ubuntu, WSL, Github